Kaixiang Yao

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Education

Southern University of Science and Technology Aug. 2023 - Jun. 2027 (Expected)
B.Eng. in Robotics Engineering; GPA: 3.91/4.00; Rank: 1/91
Advisor: Prof. Chenglong Fu

University of California, Berkeley Jun. 2026 - Aug. 2026
Visiting Undergraduate Student, Berkeley Summer Sessions; GPA: 4.00/4.00

Publications and Manuscripts

[1] Quasi-real-time digital twin for robotic assembly: a physics-based multi-fidelity framework *(Journal of Intelligent Manufacturing, 2026; JCR Q1, 5-year IF 9.2)*
Kaixiang Yao, Hao Chi, Dingye Chen, Jiwen Zhang

[2] RopeFormer: Cross-Trial Adaptation from Interaction History for Dynamic Rope Manipulation *(under review; arXiv:2609.23432, 2026)*
Menglin Wu[*], **Kaixiang Yao**[*], Shangbo Luan, Masayoshi Tomizuka, Yuxin Chen
[*] *Equal contribution.*

[3] Gaze-Driven Perception-Action Control of Multi-Channel FES for Efficient Grasp Restoration *(accepted for publication in IEEE T-MRB, 2026; JCR Q2, 5-year IF 4.3)*
Kaixiang Yao, Zeyang Yang, Jinxin Sun, Kong Hoi Cheng, Junyun Fu, Zhaokai Chen, Chengyu Lin, Chenglong Fu

Selected Technical Preprints

TOGEARI: Interaction-Space Preconditioning for Condensed Finite-Element Systems with IPC Contact *(arXiv:2608.14162, 2026)*
Yanlin Liu[*], Chao Huang[*], **Kaixiang Yao**[*], Yao Shen

RUPA: Nonlinear volume consistency, constraint geometry and singular penalty limits in finite elements *(arXiv:2609.13446, 2026)*
Yanlin Liu, Chao Huang[†], **Kaixiang Yao**[†], Yao Shen

MORTIS: Quadrature-compatible cofactor gauges and optimal material margins in finite elasticity *(arXiv:2609.13449, 2026)*
Yanlin Liu, **Kaixiang Yao**[†], Chao Huang[†], Yao Shen

[*] *Equal contribution.* [†] *Shared second authorship.*

Research Experience

University of California, Berkeley Jun. 2026 - Present
Visiting Undergraduate Researcher, Mechanical Systems Control Laboratory Advisor: Prof. Masayoshi Tomizuka

RopeFormer: Cross-Trial Adaptation from Interaction History for Dynamic Rope Manipulation

- Built the Newton-to-MJLab replay and cross-simulator validation pipeline for history-conditioned policies across dynamic rope manipulation tasks, including single-arm swing, bimanual twirl, and transient whip.
- Implemented the full Unitree H1-2 deployment stack, integrating dual ZED 2i perception, policy inference, safety constraints, and real-time control.
- Led real-world evaluation of frozen policies on unseen physical ropes across repeated trials to characterize cross-trial adaptation.

Research Experience (continued)

Shanghai Jiao Tong University Oct. 2025 - Jun. 2026
Undergraduate Researcher Advisor: Prof. Yao Shen

CARTILAGE: Robot Learning for High-Fidelity Rigid-Soft Contact (ongoing)

- Developed a simulation-and-learning framework for soft-gripper manipulation of fragile objects under high-fidelity rigid-soft contact, including pneumatic actuation, contact/tactile modeling, and RL-compatible tasks for force-aware grasping.
- Co-developed TOGEARI, an interaction-space preconditioner for condensed finite-element systems with IPC contact, motivated by solver bottlenecks encountered in CARTILAGE.

Southern University of Science and Technology Jul. 2024 - May 2026
Undergraduate Researcher Advisor: Prof. Chenglong Fu

Eye-Tracking and Electrical Stimulation-Assisted Grasping

- Developed a gaze-driven grasp-intent pipeline using normalized gaze-variance features, random-forest recognition, dual-threshold temporal filtering, YOLOv5, and MediaPipe for object-aware grasp selection.
- Designed an 8-channel, 54-electrode high-density FES sleeve and evaluated the integrated system over 67 grasping trials, achieving 90.9% task success and 98.4% visual recognition accuracy.

Tsinghua University Jun. 2025 - Aug. 2025
Undergraduate Researcher Advisor: Prof. Jiwen Zhang

Digital Twin for Robotic Assembly

- Built a ROS 2 non-blocking digital-twin pipeline that decoupled real-time safety control from background FEA, using kinematic TIE constraints and mesh-scale analysis to improve computational robustness and predictability.
- Integrated full-field stress outputs into an FSM-based safety evaluator for robotic assembly.

Technical Skills

Programming: Python, MATLAB

Robotics and Control: ROS 2, Unitree H1-2, real-time control, sim-to-real deployment

Learning and Perception: Reinforcement learning (PPO), Transformer-XL, ZED 2i stereo vision, YOLOv5, MediaPipe

Simulation and Modeling: Newton, MuJoCo/MJLab, Isaac Lab, randomized dynamics, FEA/FEM, rigid-soft contact

Mechanical Design and Fabrication: SolidWorks, 3D printing, rapid prototyping

Honors and Awards

First Prize (2nd Overall, Provincial Level) 2025
Guangdong Provincial College Students Engineering Practice and Innovation Competition

Outstanding Student Scholarship (two consecutive years) 2023 - 2025

Research Interests

My research interests are in robot learning and embodied AI for physical manipulation, especially contact-rich and deformable tasks, with a growing interest in dexterous manipulation. I am particularly interested in integrating dynamics, contact, force, and embodiment constraints with perception and closed-loop control for reliable adaptation in the physical world.